ELEC 231 — Mathematical Methods I for ECE · Fall 2026

Week 3: Techniques for first order ODEs; basic theory

Prof. Michael Korenberg · Prepared by Abdelrahman Ramadan
Typed from Prof. Korenberg's Fall 2025 handwritten notes, version 2026-09-18.

This note covers selected content from Chapters 1–3 of Murray R. Spiegel's *Applied Differential Equations*, and Chapter 7 of Nilsson & Riedel's *Electric Circuits* (12th ed.).

Notation.

j	imaginary unit, $\mathrm{j}^2=-1$	$\mathcal{L}\{f(t)\}$	Laplace transform
$\mathrm{e}^{-t/\tau}$	natural exponential	$\mathcal{L}^{-1}\{F(s)\}$	inverse transform
$\frac{\mathrm{d}y}{\mathrm{d}t}$, $\frac{\mathrm{d}^2y}{\mathrm{d}t^2}$	derivatives	$v * h$	convolution
$u(t)$	unit step	$\boldsymbol{x}$, $\mathbf{A}$	vector, matrix
$\delta(t)$	unit impulse	τ	time constant

1 Solution of a 1st order linear ODE with time-varying coefficients

Consider the first-order linear ODE

$$\frac{\mathrm{d}y(t)}{\mathrm{d}t} + a(t)\,y(t) = f(t). \tag{3.1}$$

Consider

$$y(t)\,\mathrm{e}^{\int a(t)\,\mathrm{d}t}. \tag{3.2}$$

$\mathrm{e}^{\int a(t)\,\mathrm{d}t}$ is called an *integrating factor* for Eq. (3.1), since multiplying both sides of (3.1) by this integrating factor yields

$$\mathrm{e}^{\int a(t)\,\mathrm{d}t}\frac{\mathrm{d}y(t)}{\mathrm{d}t} + a(t)\,y(t)\,\mathrm{e}^{\int a(t)\,\mathrm{d}t} = \mathrm{e}^{\int a(t)\,\mathrm{d}t} f(t). \tag{3.3}$$

The left side of (3.3) is the derivative with respect to t of expression (3.2) since

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(y(t)\,\mathrm{e}^{\int a(t)\,\mathrm{d}t}\Big) = \mathrm{e}^{\int a(t)\,\mathrm{d}t}\frac{\mathrm{d}y(t)}{\mathrm{d}t} + y(t)\,\mathrm{e}^{\int a(t)\,\mathrm{d}t}\,a(t).$$

Hence

$$\frac{\mathrm{d}\big(y(t)\,\mathrm{e}^{\int a(t)\,\mathrm{d}t}\big)}{\mathrm{d}t} = \mathrm{e}^{\int a(t)\,\mathrm{d}t} f(t), \tag{3.4}$$

so that

$$y(t)\,\mathrm{e}^{\int a(t)\,\mathrm{d}t} = \int \mathrm{e}^{\int a(t)\,\mathrm{d}t} f(t)\;\mathrm{d}t + C, \tag{3.5}$$

where C is a constant. Then $y(t)$ can be obtained using initial condition $y(0)=y_0$.

This integrating factor $\mathrm{e}^{\int a(t)\,\mathrm{d}t}$ is a function of only one variable, a special case.

Example 3.1 (Spiegel, Applied Differential Equations).

$$\frac{\mathrm{d}y(t)}{\mathrm{d}t} + \frac{10\,y(t)}{2t+5} = 10, \qquad y(0)=0.$$

Here

$$a(t) = \frac{10}{2t+5},$$

so that

$$\text{Integrating factor} = \mathrm{e}^{\int \frac{10}{2t+5}\,\mathrm{d}t} = \mathrm{e}^{\ln(2t+5)^5} = (2t+5)^5. \tag{3.6}$$

From (3.4), (3.6),

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big[\, y(t)\cdot(2t+5)^5 \,\Big] = 10\,(2t+5)^5 \qquad \textit{Eq. (3.4) with } f(t)=10$$

$$y(t)\cdot(2t+5)^5 = 10\int (2t+5)^5\,\mathrm{d}t. \qquad \textit{integrate both sides}$$

On right side, set $u = 2t+5 \Rightarrow \mathrm{d}u = 2\,\mathrm{d}t$, so

$$y(t)\cdot(2t+5)^5 = 10\int (2t+5)^5\,\mathrm{d}t = \frac{10}{2}\int u^5\,\mathrm{d}u = \frac{5}{6}(2t+5)^6 + C. \tag{3.7}$$

Since $y(0)=0$, set $t=0$ in (3.7) so

$$0 = \frac{5}{6}\,5^6 + C \implies C = -\frac{5^7}{6} = -13\,020.83,$$

so from (3.7)

$$y(t) = \frac{5}{6}(2t+5) - \frac{5^7}{6\,(2t+5)^5}.$$

1.1 An RL circuit solved with the integrating factor

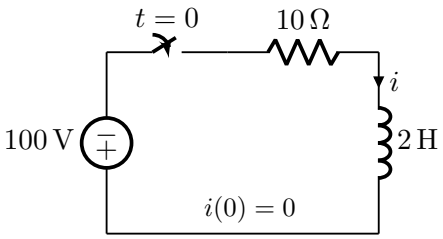


Figure 3.1. The 100 V series RL circuit of running page ④: the switch closes at $t=0$, $R = 10\,\Omega$, $L = 2\,\mathrm{H}$, and the inductor carries no current beforehand, so $i(0)=0$. The inductor voltage v is marked + at the top in the original sketch.

Applying Kirchhoff's voltage law to Figure 3.1 for $t \geq 0$:

$$L\frac{\mathrm{d}i}{\mathrm{d}t} + R\,i = 100, \qquad t \geq 0 \qquad \textit{KVL around the loop}$$

$$2\frac{\mathrm{d}i}{\mathrm{d}t} + 10\,i = 100 \implies \frac{\mathrm{d}i}{\mathrm{d}t} + 5i = 50 \qquad L = 2\,\mathrm{H},\ R = 10\,\Omega\textit{; divide by } L$$

$$\text{Integrating factor} = \mathrm{e}^{\int 5\,\mathrm{d}t} = \mathrm{e}^{5t} \qquad \textit{here } a(t) = 5 \textit{ is constant}$$

$$\mathrm{e}^{5t}\frac{\mathrm{d}i}{\mathrm{d}t} + 5\mathrm{e}^{5t}\,i = 50\,\mathrm{e}^{5t} \qquad \textit{multiply through by } \mathrm{e}^{5t}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(i\,\mathrm{e}^{5t}\right) = 50\,\mathrm{e}^{5t} \qquad \textit{the left side is an exact derivative, Eq. (3.4)}$$

$$i\,\mathrm{e}^{5t} = \int 50\,\mathrm{e}^{5t} = 10\,\mathrm{e}^{5t} + C \qquad \textit{integrate}$$

$$i(0) = 0 \implies 0 = 10 + C \implies C = -10 \qquad \textit{apply the initial condition}$$

$$i(t) = 10 - 10\,\mathrm{e}^{-5t} = 10\left(1 - \mathrm{e}^{-5t}\right)\ \mathrm{A}. \qquad \textit{divide by } \mathrm{e}^{5t}$$

2 Alternative solution to the RL circuit (separation of variables)

The same circuit, and the same ODE $\frac{\mathrm{d}i}{\mathrm{d}t} + 5i = 50$, can be solved by separating the variables:

$$\frac{\mathrm{d}i}{\mathrm{d}t} + 5i = 50$$

$$\frac{\mathrm{d}i}{50 - 5i} = \mathrm{d}t \qquad \textit{separate: all the } i \textit{ on the left, all the } t \textit{ on the right}$$

$$-\tfrac{1}{5}\ln|50 - 5i| + C_1 = t + C_2 \qquad \textit{integrate both sides}$$

$$\ln|50 - 5i| = -5t + C_3 \qquad \textit{multiply by } -5\textit{; } C_3 = 5(C_1 - C_2)$$

$$|50 - 5i| = C_4\,\mathrm{e}^{-5t} \qquad \textit{exponentiate; } C_4 = \mathrm{e}^{C_3}$$

$$50 - 5i = C_4\,\mathrm{e}^{-5t}, \qquad i < 10 \qquad 50 - 5i > 0 \textit{ while } i < 10\textit{, so the bars may be dropped}$$

$$5i = 50 - C_4\,\mathrm{e}^{-5t}, \qquad i < 10$$

$$i = 10 - C\,\mathrm{e}^{-5t} \qquad \textit{divide by } 5\textit{; } C = C_4/5$$

$$i(0) = 0 \implies 0 = 10 - C \implies C = 10$$

$$i(t) = 10 - 10\,\mathrm{e}^{-5t} = 10\left(1 - \mathrm{e}^{-5t}\right)\ \mathrm{A},$$

which is the answer already obtained with the integrating factor in Section 1.1.

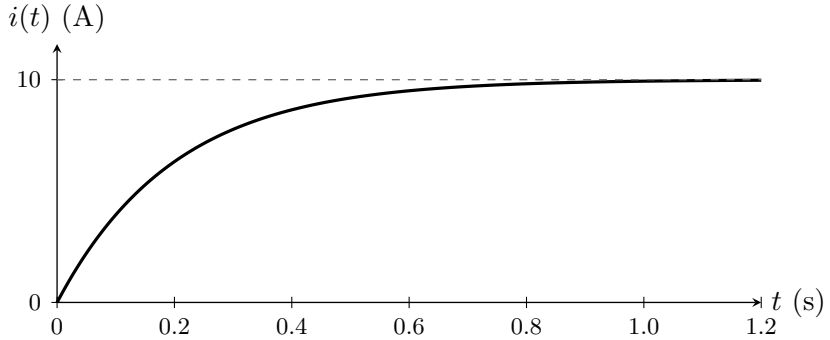


Figure 3.2. The sketch drawn beside the calculation: $i(t)$ rises from 0 and saturates at 10 A. Only the value 10 and the origin 0 are marked in the original.

Definition 3.1 (Separable ODE). A *separable* ODE has the form

$$N(y)\,\frac{\mathrm{d}y}{\mathrm{d}t} = M(t).$$

3 Solution of ODEs using separation of variables (Spiegel, Applied Differential Equations) — examples

Example 3.2 (Spiegel example (i)).

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{y^2}{t^3} \implies \frac{\mathrm{d}y}{y^2} = \frac{\mathrm{d}t}{t^3} \qquad \textit{separate}$$

$$-1\,y^{-1} = -\tfrac{1}{2}\,t^{-2} + C \implies \frac{1}{y} = \frac{1}{2t^2} - C \qquad \textit{integrate; then multiply by } -1$$

Example 3.3 (Spiegel example (ii)).

$$\frac{\mathrm{d}y}{\mathrm{d}t} = \frac{3t + t\,y^2}{y + t^2 y}, \qquad y(1) = 3$$

$$= \frac{t\,(3 + y^2)}{y\,(1 + t^2)} \implies \qquad \textit{factor } t \textit{ out of the numerator and } y \textit{ out of the denominator}$$

$$\mathrm{d}y\,\frac{y}{3 + y^2} = \mathrm{d}t\,\frac{t}{t^2 + 1} \qquad \textit{separate}$$

$$\tfrac{1}{2}\ln(3 + y^2) = \tfrac{1}{2}\ln|t^2 + 1| + c \qquad \textit{integrate; both integrands are } \tfrac{1}{2}(\textit{derivative})/(\textit{function})$$

$$3 + y^2 = C_1\,(t^2 + 1) \qquad \textit{multiply by 2 and exponentiate; } C_1 = \mathrm{e}^{2c}$$

$$C_1 = \frac{12}{2} = 6 \qquad y(1) = 3\textit{: } 3 + 3^2 = 12 \textit{ and } 1^2 + 1 = 2$$

$$y^2 + 3 = 6\,(t^2 + 1)$$

Example 3.4 (Spiegel example (iii)).

$$y\,\mathrm{d}t + (t^3 y^2 + t^3)\,\mathrm{d}y = 0$$

$$-\frac{\mathrm{d}y}{y}\,t^3\,(y^2 + 1) = \mathrm{d}t \qquad \textit{collect } t^3\textit{; move the } \mathrm{d}t \textit{ term to the right and divide by } y$$

$$-\frac{\mathrm{d}y}{y}\,(y^2 + 1) = t^{-3}\,\mathrm{d}t \qquad \textit{divide by } t^3$$

$$\int \left(y + \frac{1}{y}\right)\mathrm{d}y = -\int t^{-3}\,\mathrm{d}t \qquad (y^2 + 1)/y = y + 1/y\textit{; multiply through by } -1$$

$$\frac{y^2}{2} + \ln|y| = \frac{t^{-2}}{2} + C \qquad \textit{integrate}$$

$$y^2 + 2\ln|y| - t^{-2} = C_1 \qquad \textit{multiply by 2; } C_1 = 2C$$

4 Existence and uniqueness

Theorem 3.2 (Existence and Uniqueness Theorem). Given the 1st order differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}t} = F(t, y),$$

let $F(t, y)$ satisfy the following conditions:

1. $F(t, y)$ is real, finite, one-valued, and continuous at all points (t, y) within a region R of the ty plane (which may comprise all points).
2. $\dfrac{\partial F(t, y)}{\partial y}$ is real, finite, one-valued and continuous in R.

Then there exists *one and only one* solution $y = g(t)$ in R, which passes through any given point of R.

Source: Spiegel, *Applied Differential Equations*, p. 21.

Spiegel points out that this theorem provides *sufficient* conditions for existence and uniqueness of a solution, but the conditions are NOT necessary.